D 13145 **(Pages : 3)** **Name.....**

Reg. No.....

FIRST SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, NOVEMBER 2021

[November 2020 session for SDE/Private students]

(CBCSS)

Mathematics

MTH1C03—REAL ANALYSIS—I

(2019 Admission onwards)

{Covid instructions are not applicable for Pvt/SDE students (November 2020 session)}

Time : Three Hours Maximum : 30 Weightage

General Instructions

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section / Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section / sub part / sub division may be ignored.*
4. *There will be an overall ceiling for each Section / Part that is equivalent to the maximum weightage of the Section / Part.*

Part A (Short Answer Questions)

Answer ***all*** *questions.*
Each question carries 1 *weightage.*

1. Show that a finite point set has no limit points.
2. If f is monotonic on (a, b), prove that the set of discontinuities of f on (a, b) is at most countable.
3. Suppose f is a real differentiable function on $[a, b]$ and suppose $f'(a) < \lambda < f'(b)$. Prove that there is a point $x \in (a, b)$ such that $f'(x) = \lambda$.
4. If $f \in \mathcal{R}(\alpha)$ on $[a, b]$, prove that $|f| \in \mathcal{R}(\alpha)$ and $\left|\int_a^b f \, d\alpha\right| \leq |f| \, d\alpha$.

Turn over

5. If $a < s < b$, f is bounded on $[a, b]$, f is continuous at s, and $\alpha(x) = l(x - s)$, then prove that $\int_a^b f \, d\alpha = f(s)$.

6. State and prove the integration by parts rule.

7. Show by an example that the process of taking limits can not be interchanged in general.

8. Define pointwise boundedness and uniform boundedness of a sequence of functions.

(8 × 1 = 8 weightage)

Part B

Answer any **two** *questions of each unit.*
Each question has weightage 2.

UNIT I

9. If $\{E_n\}$, n = 1, 2, 3,... is sequence of countable sets and if $S = \bigcup_{n=1}^{\infty} E_n$. then prove that S is countable.

10. Prove that the Cantor set is a perfect set.

11. Let f be a real uniformly continuous function on the bounded set E in R^1. Prove that f is bounded on E.

UNIT II

12. Let f be defined on $[a, b]$. If f has a local maximum at a point $x \in (a, b)$, and if $f'(x)$ exists, then prove that $f'(x) = 0$.

13. If $f(x) = |x|^3$, compute $f'(x), f''(x)$ for all real x, and show that $f^{(3)}(0)$ does not exist.

14. State and prove a necessary and sufficient condition for Riemann-Stieltjes integrability.

UNIT III

15. State and prove the fundamental theorem of calculus for vector valued functions.

16. Suppose $f_n \to f$ uniformly on a set E in a metric space. Let x be a limit of E and suppose that $\lim_{t \to x} f_n(t) = A_n \; (n = 1, 2, 3, \ldots)$. Then prove that $\{A_n\}$ converges and $\lim_{n \to \infty} A_n$.

17. If K is a compact metric space, if $f_n \in \mathcal{C}(K)$ for n = 1, 2, 3,..., and if $\{f_n\}$ converges uniformly on K, prove that $\{f_n\}$ is equicontinuous on K.

(6 × 2 = 12 weightage)

Part C

Answer any **two** *from the following four questions (18-21).*
Each question has weightage 5.

18. (a) Let X be an infinite set. For $p \in X$ and $q \in X$, define :

$$d(p, q) = \begin{cases} 1 \ (if \ p \neq q) \\ 0 \ (if \ p = q) \end{cases}$$

Prove that this is a metric. Which subsets of the resulting metric space are open ? Which are closed ? Which are compact ?

(b) State and prove a necessary and sufficient condition for continuity of a function using open sets.

19. (a) State and prove the L′ Hospital's rule.

(b) Prove that if f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.

20. (a) State and prove the Cauchy's criterion for uniform convergence of a sequence of functions.

(b) Suppose $\{f_n\}$ is a sequence of functions, differentiable on $[a, b]$ and such that $\{f_n(x_0)\}$ converges for some point x_0 on $[a, b]$. If $\{f_n'\}$ converges uniformly on $[a, b]$, prove that $\{f_n\}$ converges uniformly on $[a, b]$, to a function f, and $f'(x) = \lim_{n \to \infty} f_n'(x) \ (a < x < b)$.

21. State and prove the Weierstrass's approximation theorem. Hence show that for every interval $[-a, a]$, there is a sequence of real polynomials P_n such that $P_n(0) = 0$ and such that $\lim_{n \to \infty} P_n(x) = |x|$ uniformly on $[-a, a]$.

(2 × 5 = 10 weightage)